Iron ore into China: delivered cost and the import window

Nathan Couturier. Research note, data to 30 Oct 2026 (daily) and Sep 2026 (monthly benchmark). Interactive version, data and code: https://nathancouturier.github.io/iron-ore-china-arbitrage/

Question

When does it pay to bring a cargo of seaborne iron ore into China, from where, and who captures the margin between the mine and the blast furnace? The note traces the delivered cost of a 62% Fe cargo along the voyage: the index-equivalent FOB netback (the CFR index price minus modeled Capesize freight), CFR Qingdao, then USD/CNY, import VAT, port charges and voyage financing to the landed cost, which it compares with the Dalian futures price for the matching delivery; then it asks whether Chinese steel mill margins support the ore price.

Answer

- On 30 Oct 2026 a 62% Fe cargo from West Australia bought for delivery to Qingdao in about 45 days cost 739 RMB per dry tonne landed, VAT included, against 692 for 45-day Dalian delivery: a margin of −47 RMB/dmt, so the window is closed. Without the 2.73 USD/dmt lift that puts 2026 SGX prices on a 62% Fe basis the margin would be −26, still below the band, and a CFR price 2.9 USD lower would bring it inside the band.
- The window has almost never been open. On the daily series (from December 2015) the margin exceeded the 25 RMB/dmt cost of using the exchange on 0.9% of trading days and never for more than 6 days in a row; on the monthly baseline it was below the band in 97% of 153 months and averaged −79 RMB/dmt (9.6% of landed cost).
- Gaps fade slowly: the half-life is 5.0 months on monthly data (95% interval 2.2 to 8.1), but a unit root cannot be rejected there (ADF p = 0.16); on daily data the half-life is 23 trading days (ADF p = 0.010). The margin reverts to a negative level, not to zero.
- Freight takes 9.0% of the CFR price from West Australia and 21.5% from Brazil on average since 2014; Brazil's index-equivalent netback trails Australia's by 12.1 USD/dmt on average and by 29.2 in Sep 2026.
- Most of the discount sits in Dalian. On 5 dated Fastmarkets Qingdao port prices (2019 to 2022), Dalian's 45-day price was 72 to 374 RMB/dmt below the port price of 62% fines, while the port price minus the landed cost ranged from −88 to +81 RMB/dmt. Mills are squeezed: the rebar futures margin was −79 RMB/t on 2 Nov 2026, negative on every trading day of 2026, and the pre-registered test finds no sign that low mill margins precede weaker ore.

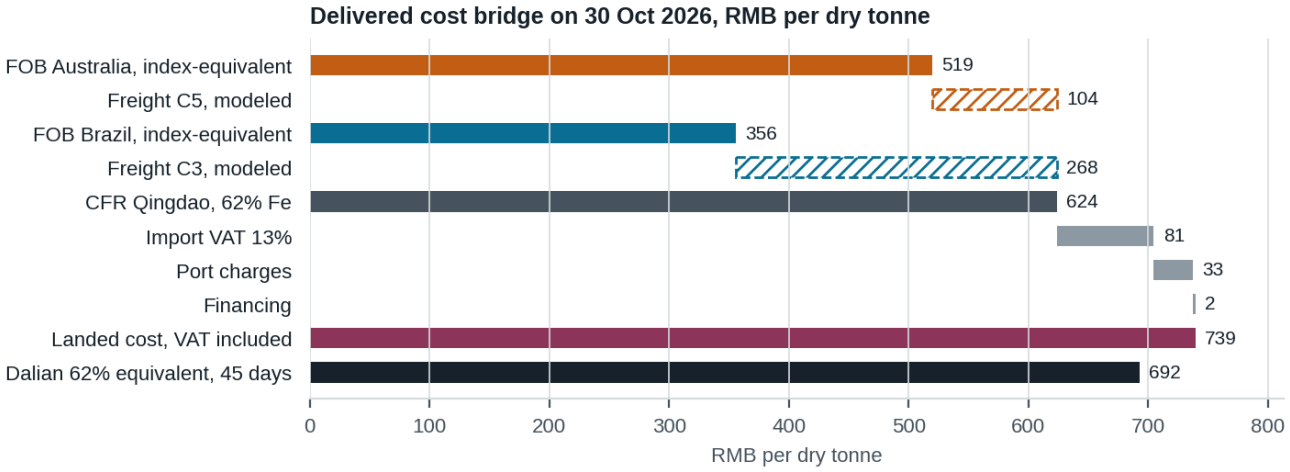


Figure 1. Freight is modeled (hatched). FOB values are index-equivalent netbacks (CFR minus freight per dry tonne) and ignore brand premiums and discounts. Sources: SGX, Dalian Commodity Exchange via Sina, FRED; freight modeled from the Baltic Capesize Index and Brent.

Data and conventions

Seaborne price: IMF 62% Fe CFR China monthly average (FRED PIORECRUSDM) for the monthly baseline from January 2014; SGX iron ore futures interpolated to a 45-day horizon for the daily series from December 2015 (the current-month contract is not used because it settles on the month's average). SGX contracts from January 2026 settle on Platts IODEX at 61% Fe and are lifted to 62% by one constant, the mean measured monthly spread to the IMF 62% average (2.73 USD/dmt). The daily series compares forward with forward (the hedgeable trade); the monthly baseline compares the spot index with 45-day Dalian and shows a deeper discount in steep backwardation. Dalian: every iron ore contract from the exchange's annual files (2013 to 2024) and Sina (2025 onward), expressed on a 62% Fe basis (standard grade 61% from I2205, plus 10 x X with

X = 1.5 RMB per 0.1% Fe in every period), interpolated to a constant 45-day maturity across contracts with at least 1,700 lots of open interest. USD/CNY from FRED (DEXCHUS); SOFR (fed funds before April 2018) plus 150 bp for financing; import VAT 17%, 16% from 1 May 2018 and 13% from 1 April 2019, duty zero. Port charges 30 RMB per wet tonne and moisture 8% (per dry = per wet / 0.92). Rebar and hot-rolled coil from the Shanghai Futures Exchange; coke from Dalian.

The import window

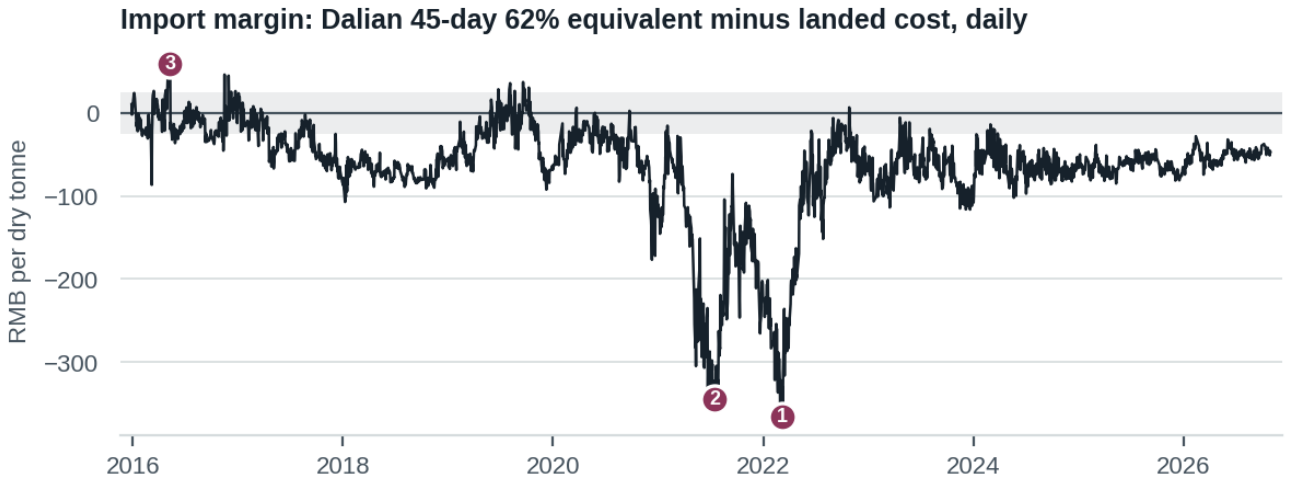


Figure 2. Shaded: the no-arbitrage band of ±25 RMB/dmt (delivery fee, about a month of warehouse storage and the vessel outbound fee cap at the fees in force since mid-2021; they were higher before). Numbered: the three largest deviations from the mean, at least 120 trading days apart. Sources: SGX, Dalian Commodity Exchange, FRED; computed by the pipeline.

	Monthly baseline	Daily series
Period	Jan 2014 to Sep 2026	28 Dec 2015 to 30 Oct 2026
Observations	153	2,613
Mean, median (RMB/dmt)	−79.4, −69.3	−66.8, −57.1
Share open, inside, closed	0.0%, 3.3%, 96.7%	0.9%, 18.9%, 80.2%
Minimum	−408.9 (Jul 2021)	−366.2 (7 Mar 2022)
AR(1) half-life (95% interval)	5.0 months (2.2 to 8.1)	22.6 days (16.0 to 29.5)
ADF statistic, p-value	−2.33, 0.161	−3.43, 0.010

Table 1. Import margin statistics. Open: above +25 RMB/dmt; closed: below −25.

The two largest deviations have documented policy causes; the third is an artifact of an expiring contract. (1) 7 Mar 2022, −366 RMB/dmt: Chinese regulators warned against iron ore speculation and port hoarding in February 2022, weighing on Dalian, while seaborne prices rose on supply losses in Australia and Brazil and the war in Ukraine. [NDRC price department, February 2022; NDRC and SAMR meeting with port operators, February 2022; World Bank, Commodity Markets Outlook, April 2022, p. 53] (2) 15 Jul 2021, −344 RMB/dmt: China ordered 2021 crude steel output to fall year on year. Dalian priced the cuts while seaborne 62% stayed above 200 USD/dmt; the World Bank attributes the 2021Q3 fall in iron ore prices to them. [NDRC and MIIT, 1 April 2021; World Bank, Commodity Markets Outlook, October 2021, p. 9] (3) 9 May 2016, +59 RMB/dmt: Dalian's 45-day price sat above import parity from 3 to 10 May 2016, but the reading comes from the expiring May contract (I1605, 7 days to expiry, about 3,400 lots open against 489,000 in September), which settled about 107 RMB above September; without delivery-month contracts the run disappears. The exchange had raised iron ore fees twice (to 0.9, then 1.8 per 10,000 of turnover) and the minimum margin to 8% during the April futures rally. [DCE notice, 21 April 2016 (fees); DCE notice, 25 April 2016 (fees); DCE notice, 21 April 2016 (margins)]

Sensitivity of the latest monthly margin (Sep 2026, 78.7 RMB/dmt below the open threshold): Port charges (20 / 40 RMB/wmt) +10.9 / −10.9; No-arbitrage band (15 / 35 RMB/dmt) +10.0 / −10.0; FX source (PBoC central parity) −6.2; Maturity target (30 / 90 days) +1.3 / −5.1; Route financing (Brazil, C3) −2.6. No single assumption in its range opens the window. Against the most-active contract (about 108 days to expiry on average) the mean monthly margin would be −104.3 instead of −79.4 RMB/dmt, because the Dalian curve is usually backwardated.

Imports (pre-registered; China Customs monthly bulletins from 2018): a monthly margin 50 RMB/dmt higher precedes +3.1 percentage points of year-on-year import growth over the next 3 months (95% interval +2.3 to +3.9, Newey-West t 7.7, 87 months). The link rests on 2021 and 2022, when policy pushed Dalian far below import parity while imports fell; without those years (an exploratory check) the slope is +2.5 with an interval from −3.2 to +8.1. The data do not separate an effect of the window on imports from the policy that moved both.

Freight and origins

Capesize route rates (C3 Tubarao to Qingdao, C5 West Australia to Qingdao) are paid data, so freight is modeled. A voyage model converts the Baltic Capesize Index into a time charter rate (calibrated by regime on 62 dated public pairs) and Brent into a bunker price (calibrated separately before and after the IMO 2020 sulphur cap), then prices the round voyage at the Baltic eco speed of 12 knots over searoute distances (3,531 and 11,137 nautical miles). It is never fitted to a route rate; against published assessments it scores RMSE 1.04 USD/t on C5 (n = 91) and 2.04 on C3 (n = 90); a regression on the BCI and Brent scores 0.92 and 1.65 out of sample. In the voyage model the C3 minus C5 spread rises by about 3.3 USD/t per 1,000 BCI points (correlation 0.91). In 2026 freight takes 13.6% of CFR from Australia and 34.4% from Brazil, the highest annual shares since 2014.

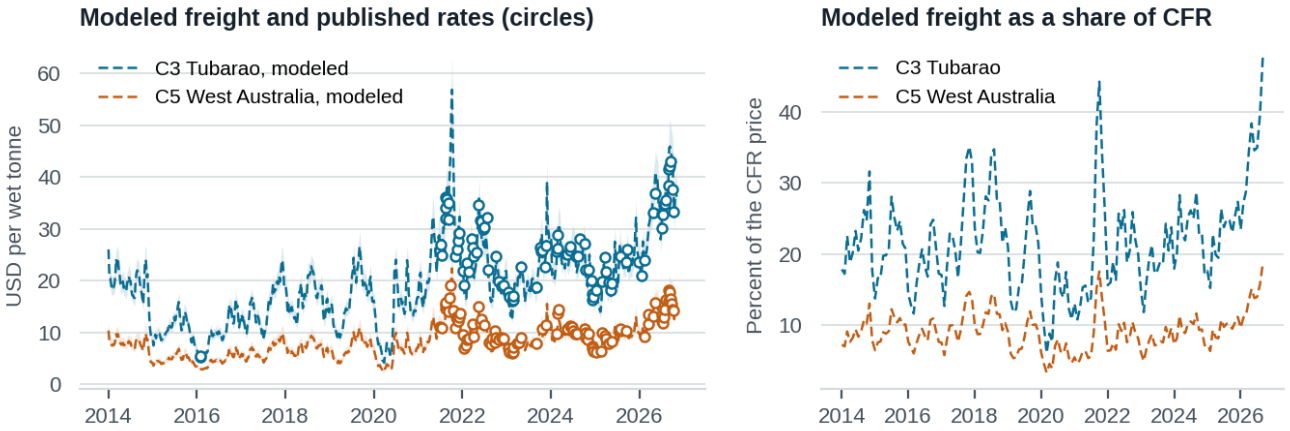


Figure 3. Left: modeled C3 and C5 (dashed lines) with shaded 80% error bands and the published assessments used to score them (circles: Baltic Exchange via the shipping press, Platts, Ruida Futures). Right: modeled freight per dry tonne as a share of the 62% CFR price. Sources: East Money (BCI), FRED, IMF.

Mill margins

The Chinese market convention values a tonne of steel against 1.6 t of iron ore, 0.5 t of coke and a fixed 1,080 RMB/t of other costs (rebar; HRC 150 more), a broker convention rather than an accounting margin. The pre-registered test (written before computing it) asks whether a rebar margin in the bottom decile of its previous three years precedes weaker Dalian ore over one to three months.



Figure 4. Most-active contracts. Sources: Shanghai Futures Exchange, Dalian Commodity Exchange, Sina.

Margin	Horizon	Signal days	Effect, pp	95% interval	Newey-West t
Rebar	21 days	581	+0.94	−1.9 to +4.1	0.66
Rebar	42 days	567	+3.83	−1.2 to +9.5	1.44
Rebar	63 days	549	+5.68	−1.2 to +13.1	1.51
HRC	21 days	358	+1.52	−1.2 to +4.6	1.07
HRC	42 days	347	+5.00	+0.5 to +10.4	2.12
HRC	63 days	340	+5.37	−1.2 to +12.7	1.55

Table 2. Difference in the forward log change of the Dalian 45-day price between signal days and other days; 95% intervals from a moving block bootstrap (63-day blocks). The hypothesis was a negative effect; every estimate is positive and only HRC at 42 days has an interval that excludes zero.

The long view

The margin needs Dalian futures, listed in October 2013; the seaborne price, delivered cost and China's imports go back further (price and imports in Figure 5). In dollars of Sep 2026, a tonne of 62% ore averaged 55 USD from 1960 to 2003 under annual contracts FOB Brazil, against 104 over the last twelve months CFR China (the change of basis in 2006 adds the sea freight); the real peak was 304 in Nov 2007. The delivered cost in China peaked at 1,635 RMB per dry tonne in Nov 2007, when import VAT was 13%, and was 775 in Sep 2026. China's imports grew from 25 Mt in 1992 to 1,260 Mt in 2025: Australia's share went from 55% to 60%, Brazil's from 19% to 22%, and India's peaked at 25% in 2005 before falling to 2%.

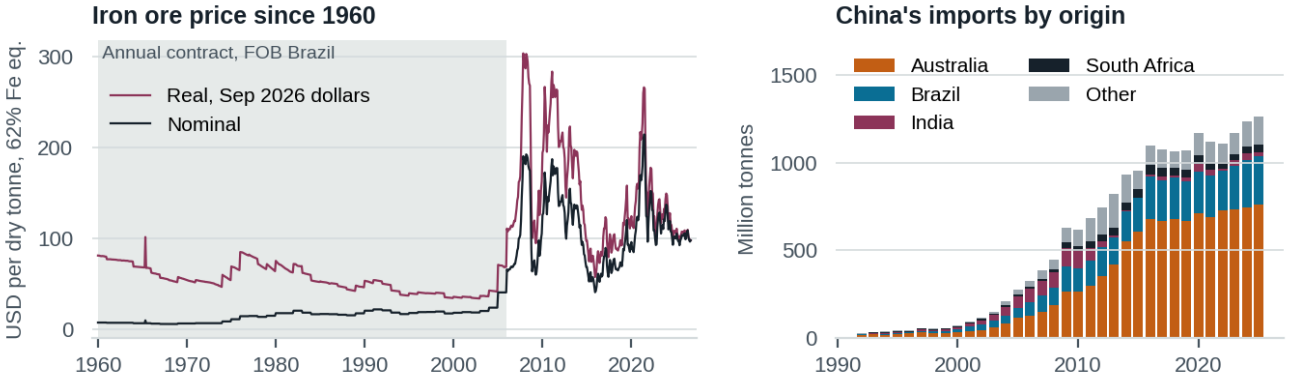


Figure 5. Left: World Bank iron ore price as a 62% Fe equivalent, annual contracts FOB Brazil quoted per Fe unit (shaded, to 2005) and spot CFR China from 2006, nominal and deflated by US CPI-U; May 1965 is shown as published. Right: China's iron ore imports by origin, HS 260111 and 260112, UN Comtrade via World Bank WITS (within 1% of the China Customs totals from 2018).

Validation

Check	n	Result	Tolerance
Import cost, Ruida Futures estimates, FX that best fits theirs (PBoC central parity)	8	max error 6.3 RMB/t (22.2 with the New York rate used for the headline)	15 RMB/t
Fastmarkets Qingdao port price to USD/dmt	5	max error 1.51 USD/dmt	2 USD/dmt
Rebar margin, SDIC Essence Futures (independent)	1	error −0.01 RMB/t	5 RMB/t
Rebar margin, Everbright Futures 2026 (constant calibrated on these)	5	max error 4.4 RMB/t; checks prices and contracts only	calibration
C3 freight, voyage model	90	RMSE 2.04 USD/t	3.0 USD/t
C5 freight, voyage model	91	RMSE 1.04 USD/t	1.5 USD/t

Table 3. Tolerances are enforced by the test suite. The mill margin's other-cost constant is calibrated on Everbright Futures, so those rows are not independent checks; the 30 RMB/wmt port charge was set with the Ruida and Fastmarkets figures in view (Ruida's imply an adder of 30 to 35 RMB/t over goods and VAT, Fastmarkets' a port charge of 31 to 40 RMB/wmt), so those two rows test the price, FX, VAT and moisture conventions more than the port charge. The freight model's time charter and bunker maps are fitted on separate public pairs, never on route rates, so the C3 and C5 rows are out of sample. Each anchor, with its link, is listed on the Method and data page.

Limitations

The margin is an expected margin, not a riskless arbitrage: price, basis, grade and currency risk remain while the cargo is at sea. Dalian prices the cheapest deliverable brand with fixed quality discounts, so part of the persistent discount is a brand and grade effect. Freight is modeled, with the errors shown above; the published anchors cluster in 2021 to 2026 and the model has run above the most recent C3 rates (mean error +4.1 USD/t on the last five). Port charges are a single assumption. The daily series uses SGX futures, lifted from a 61% to a 62% basis in 2026 by one constant, which moves the latest margin by about 21 RMB/dmt; against the World Bank series instead of the IMF one the 2026 spread averages −0.16 USD, which would put the margin at −25. The third episode rests on an expiring contract. Import volumes come from the English China Customs bulletins, which start in 2018, and the imports test cannot separate an effect of the window from the policy episodes of 2021 and 2022.

Sources

FRED: IMF iron ore (PIORECRUSDM), Federal Reserve USD/CNY (DEXCHUS), New York Fed SOFR, effective federal funds rate (DFF), EIA Brent (DCOILBRENTEU), BLS US CPI-U (CPIAUCSL); World Bank Pink Sheet; UN Comtrade via World Bank WITS; Dalian Commodity Exchange annual files and rules; Shanghai Futures Exchange daily files; Sina futures history; SGX daily settlement files; Baltic Capesize Index via East Money; SAFE central parity; China Customs (GACC) monthly bulletins; searoute; Natural Earth. Validation: Ruida Futures daily reports, Fastmarkets, Everbright Futures, SDIC Essence Futures, Baltic Exchange route reports in the shipping press, Platts route assessments. Episode sources: NDRC, MIIT, World Bank Commodity Markets Outlook, DCE notices. Data sources with links and retrieval dates, and every validation anchor with its link: https://nathancouturier.github.io/iron-ore-china-arbitrage/#method. Code: https://github.com/nathancouturier/iron-ore-china-arbitrage.